Université Catholique d'Afrique Centrale (UCAC)

Institut Catholique de Yaoundé (ICY)

COURSE TITLE: MATHEMATICAL ANALYSIS
DEPARTMENTS: BME1
LECTURER: ARREY Marinus
INSTRUCTIONS: ANSWER ALL QUESTIONS

RESIT EXAMINATION June 2026

Question 1: (4marks)

a) Solve for x the equation $4^x - 3(2^{x-1}) + 9 = 0$

b) $\text{Log}_2(X+2) + \text{Log}_2(X+4) = 1$

Question 2 (8 marks)

a) Using the Lagrange multiplier solve the following total profit-maximizing function of a firm given as $f(x,y) = 80X - 2X^2 - XY - 3Y^2 + 100Y$ and its maximum output capacity given as follows $g(x,y) = X+Y+12$

b) Interpret the Lagrange multiplier.

c) Verify whether it is a minimal or a maxima

Question 3: (4marks)

The production function of a commodity is: $Q = 10L - 0.1L^2 + 15K - 0.2K^2 + 2KL$ where L is labour, K is capital and Q is production.

a) Calculate the marginal products of the two inputs when 10 units of each labour and capital are used,

b) If 10 units of capital are used, what is the upper limit for the use of labour which a rational producer will never exceeds.

c) Determine the marginal rate of technical substitution (MRTS) for LK and for KL

Question 4: (4marks

Find the consumer and the producer surplus given the demand function $4P = 60 - 8Q$, and the supply function $2P = 8 + Q$.

Good Luck!